

Title: Automatic News Summarization Using Large Language Models

Abstract

The rapid expansion of online news platforms has led to an overwhelming volume of information, making it increasingly difficult for readers to quickly identify relevant and reliable content. Automatic News Summarization has emerged as an effective Natural Language Processing (NLP) solution that condenses lengthy news articles into concise, informative summaries while preserving their essential meaning and factual accuracy. This project presents a comparative study of prominent transformer-based summarization models, namely Text-to-Text Transfer Transformer (T5), Bidirectional and Auto-Regressive Transformers (BART), Pegasus, and a modern Large Language Model (LLM), to evaluate their effectiveness in abstractive news summarization.

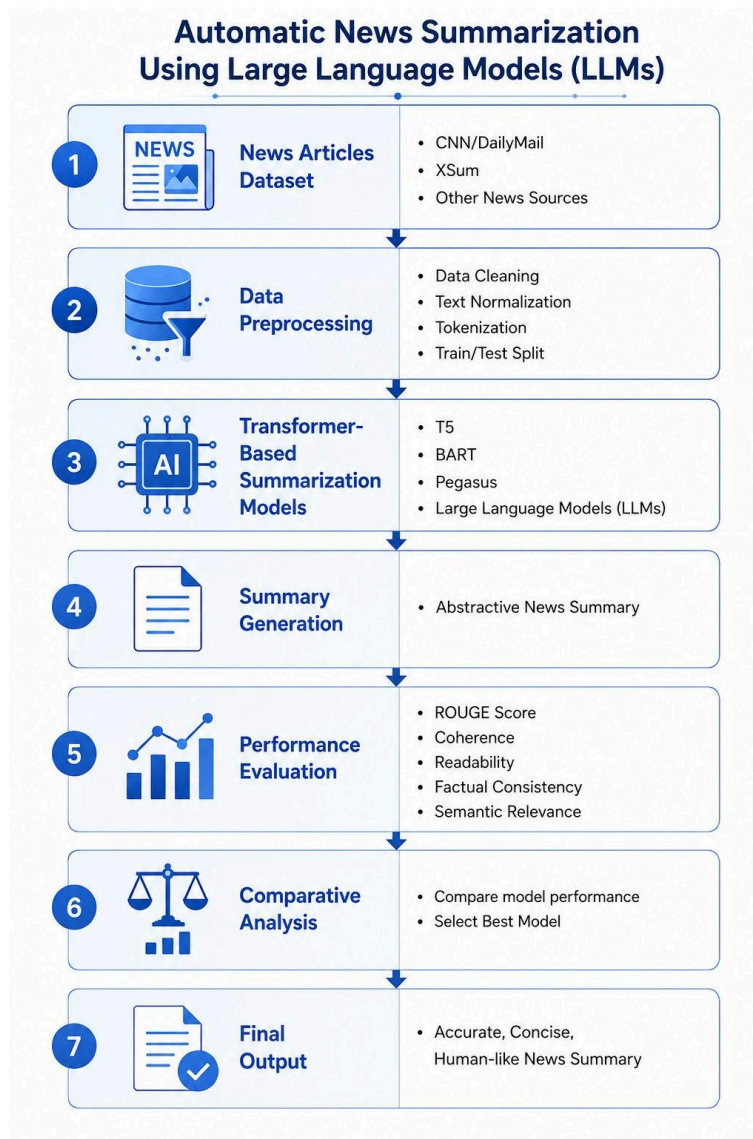
The models are trained and evaluated using two widely recognized benchmark datasets: CNN/DailyMail, which contains multi-sentence summaries of detailed news articles, and XSum, which focuses on generating highly abstractive single-sentence summaries. Performance is quantitatively assessed using standard evaluation metrics such as ROUGE-1, ROUGE-2, and ROUGE-L, while qualitative analysis examines coherence, fluency, factual consistency, semantic relevance, and readability. Comparative experiments are conducted to identify the strengths, limitations, and trade-offs of each architecture in terms of summary quality, computational efficiency, and generalization across different news domains.

The study provides a comprehensive evaluation of contemporary transformer-based and large language models for automatic news summarization and identifies the most suitable approach for producing accurate, coherent, and human-like summaries. The findings demonstrate the growing potential of advanced language models to improve information accessibility, reduce reading time, and support intelligent content management systems for real-world news applications.

Datasets

Dataset	URL link
CNN/DailyMail	https://www.kaggle.com/datasets/gowrishankarp/newspaper-text-summarization-cnn-dailymail
XSum(Extreme Summarization)	https://huggingface.co/datasets/EdinburghNLP/xsum

Approach



Project Group Members

S.No.	Roll.No.	Student Name
1	2420030187	G. Venkata Varun
2	2420030111	Srikar Jonnalagadda
3	2420030228	K. Pranav

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